HGFX: reference-compatible hierarchical Gaussian filtering in Python

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Abstract

HGFX provides hierarchical Gaussian filtering in Python against the frozen MATLAB HGF Toolbox 8.2.0 reference. We evaluate version 1.0.0 at three levels: agreement of fixed-parameter calculations, agreement of fitted model selections, and recovery of known parameters and generating models. On two published behavioral input files, the same fixed-parameter perceptual model agreed across implementations in all 324 source-defined 80-trial blocks (25,920 trials). No array element exceeded the prespecified tolerance across 12 outputs per block. Python emitted warnings in 48 blocks. The comparison covers the forward filter at one parameter vector. The original studies' observation models, parameter estimates and substantive results remain unevaluated. In 36 simulated model-selection datasets, both implementations selected the same minimum Bayesian information criterion model, with generating-model balanced accuracy of 0.583 in each. Parameter recovery fell short of some predeclared criteria, and a separate trial-horizon experiment failed its integrity gate. A continuous-input worked example illustrates the workflow on 614 empirical exchange-rate observations. These results document where the released implementation reproduces the reference and where model recovery and fitting still limit interpretation.

1 Introduction

A researcher moving a learning analysis from MATLAB to Python needs more than a function with a familiar model name. Existing inputs, parameter priors, trial exclusions and output fields must retain their meaning. The practical objective is to integrate an established analysis into another computing environment without silently changing the analysis itself. When the analysis includes a recursive Bayesian filter and nonlinear parameter estimation, verifying that objective requires comparisons at several stages. A matching final plot can conceal differences in responses, priors or optimization; a matching equation can coexist with different fitted solutions.

The Hierarchical Gaussian Filter (HGF) describes learning about states whose uncertainty and volatility change over time (Mathys et al. 2011; Mathys et al. 2014). It combines a hierarchical generative model with approximate inference that updates beliefs after each observation. Applications include studies of prediction errors and social learning (Iglesias et al. 2013; Diaconescu et al. 2014). The MATLAB implementation associated with TAPAS supplies model configurations, fitting,

simulation and diagnostic workflows (Frässle et al. 2021). These supporting conventions matter to an applied analysis because they determine which quantities are estimated and how the resulting evidence is calculated.

For example, an unconstrained optimization parameter may represent the logarithm of a positive native parameter. A prior on the former is not a Gaussian prior on the latter. A zero prior variance can mean that a parameter is fixed, while a NaN entry can be a structural placeholder rather than missing participant information. Missing responses need not imply missing environmental inputs: a participant can fail to respond on a trial that still changes the learner's beliefs. Replacing any of these conventions with a convenient default can produce an executable analysis with a different statistical interpretation.

Numerical evaluation adds another layer. The HGF is recursive, so a local discrepancy can propagate through later beliefs. Fitting adds derivatives, optimizer stopping decisions and curvature calculations. Agreement of a fixed-parameter trajectory is therefore necessary evidence for a fitting comparison, but it does not imply identical fitted parameters. Conversely, a fitted endpoint discrepancy does not locate the underlying cause. It may reflect a changed observation model, a different transform, a numerical derivative, or sensitivity within the reference implementation. These possibilities require distinct checks rather than a single correlation between output vectors. General guidance on computational modeling likewise distinguishes implementation checks from recovery and substantive inference (Wilson and Collins 2019), within the broader requirement for computational reproducibility (Peng 2011).

Python implementations already address HGF-related modeling. In particular, pyhgf supports predictive-coding models through configurable networks and differentiable computation (Legrand et al. 2026). Its availability makes a first-implementation claim inappropriate. HGFX addresses a narrower need: preserving the tested behavior of one fixed MATLAB reference, including conventions that existing workflows rely on. It is complementary to a framework whose design emphasizes flexible model construction. A useful comparison must identify a common statistical and numerical scope before assigning meaning to differences. Unsupported comparisons should remain unresolved rather than become rankings of entire packages.

This article asks how an established reference HGF workflow can be transferred to Python while preserving model semantics and the known limits of inference. The evaluated release is HGFX 1.0.0. Features added on the active development line are outside this evidence scope. The reference is HGF Toolbox 8.2.0 at a recorded immutable commit, rather than a moving upstream branch. This choice makes the target inspectable and allows later users to understand which behavior was reproduced. It also limits generalization to other toolbox releases and customized analysis pipelines.

HGFX supplies a MATLAB-independent workflow for fitting, simulation and sampling with a documented compatibility contract. The evaluation tests fixed-parameter perceptual calculations on two published behavioral input files and separates that agreement from fitting and model recovery in simulations. A continuous-input worked example illustrates the user workflow. Source audits identify additional requirements for transporting customized participant analyses.

2 Statistical formulation

2.1 Perceptual states and responses

Let u_t be an environmental input and y_t a participant response on trial t. The perceptual model updates beliefs about hidden states given the input history; the response model assigns probability

to responses conditional on those beliefs. These roles must remain separate even when both input and response are binary. An environmental outcome is not interchangeable with a participant choice.

For the binary hierarchy, a schematic generative representation is

$$p(u_t = 1 \mid x_{2,t}) = s(\kappa_1 x_{2,t}), \qquad s(a) = (1 + e^{-a})^{-1},$$

$$x_{i,t} \mid x_{i,t-1}, x_{i+1,t} \sim \mathcal{N}(x_{i,t-1} + \rho_i \Delta_t, \Delta_t \exp(\kappa_i x_{i+1,t} + \omega_i)), \quad 2 \le i < L.$$

At the top level, the innovation variance is $\Delta_t \exp(\omega_L)$ and the drift convention is retained. This summarizes the hierarchical model; it is not a claim that the implemented approximate filter samples these latent transitions. In the MATLAB binary representation, level 1 stores quantities associated with the binary outcome and the continuous hidden hierarchy begins at level 2. Treating these indices as an extra independent Gaussian outcome state would change the model (Mathys et al. 2011; Mathys et al. 2014).

The recursion stores predicted and posterior means and variances. In the frozen interface, `infStates` has trial, level and quantity axes; the quantity axis contains predicted mean, predicted variance, posterior mean and posterior variance in that order. Classic, enhanced and unbounded HGF variants differ in their update rules. Sharing a generative description does not make their approximate updates interchangeable. The source audit and component fixtures, rather than this abbreviated formulation, define their executable semantics.

For illustration, a unit-square sigmoid response model converts a predicted binary belief q_t into a choice probability,

$$\pi_t = \frac{q_t^\zeta}{q_t^\zeta + (1 - q_t)^\zeta}, \qquad \zeta > 0,$$

with Bernoulli log likelihood $y_t \log \pi_t + (1 - y_t) \log(1 - \pi_t)$. The reference evaluates this with specific logarithmic expressions and boundary handling. Algebraic equivalence does not authorize arbitrary clipping or replacement with a numerically different formula. The reference response's returned prediction and standardized residual are based on the selected belief quantity; they should not automatically be interpreted as π_t and an ordinary unstandardized response error. Prediction-versus-posterior configuration is also part of the model contract.

2.2 Parameter spaces and estimation

Write z for transformed parameters and $\theta = T(z)$ for native parameters. Positive variances and couplings use the configured transformations, including exponentiation of log parameters. The standard binary perceptual order groups initial means, log initial variances, drifts, log couplings and tonic log volatilities. Specialized families append their documented parameters. Parameter order is preserved through configuration, fitting, simulation and result export. Users should inspect named parameter fields alongside flattened vectors.

Gaussian priors are defined in the configured transformed space. Let F index free parameters with nonzero, non-NaN prior variances v_j, and let m_j denote prior means. The fitted objective is

$$\mathcal{J}(z) = -\sum_{t \in R} \log p(y_t \mid u_{1:t}, T(z)) + \frac{1}{2} \sum_{j \in F} \left[\log(2\pi v_j) + \frac{(z_j - m_j)^2}{v_j} \right].$$

Here R is the set of response trials retained by the reference likelihood contract. Fixed entries are excluded from the prior sum and optimization; structural NaN placeholders are not estimated as missing parameters. The finite prior normalizing constants are retained because likelihood and posterior density have different meanings. A change in transformed-space prior is a change in statistical specification even if native starting values look similar.

Belief inference and parameter optimization solve different problems. The filter produces a trajectory conditional on parameters; maximum a posteriori optimization searches for parameters using the likelihood and priors. Curvature at the fitted solution then supports the documented approximate covariance and Laplace evidence calculations. Bayesian information criterion (BIC) uses the negative log likelihood, free-parameter count and reference observation count; it is not the negative log joint. The evaluation consequently checks objective components and statistical outputs separately. No new inferential method is proposed here.

2.3 Missing observations and time

HGFX preserves the reference distinction between irregular responses and ignored inputs. An absent response can be excluded from the observation likelihood while its input remains available to the perceptual recursion. An ignored environmental input has different implications for state updating. The public result exposes the relevant metadata, including MATLAB-style indexing where that interface is requested. Users should not remove input rows merely to make response vectors complete.

Non-unit trial intervals require the corresponding configuration, with intervals in the documented input column. Appending a time column alone is insufficient evidence that the filter uses it. The same applies to additional input columns used by specialized models. Input shape, coding, masks and configuration must be established together. These details are especially important for transporting custom studies: a default binary model cannot infer an unpublished experiment's coding conventions.

3 Software and workflow

3.1 Installation and analysis contract

The evaluated Python package is installed with `pip install hgfx==1.0.0`; its source release is pinned in the supplement. HGFX uses NumPy and JAX (Bradbury et al. 2018; Frostig et al. 2018). User analyses do not require MATLAB. MATLAB is needed to regenerate oracle outputs, which is a different activity from executing a fitted analysis in Python. Repository demo exporters also need reference-source inputs that are not ordinary wheel resources.

The public surface provides Python-first functions `fit_model`, `sim_model` and `sample_model`, alongside MATLAB-style aliases. A fit takes aligned responses and inputs plus perceptual, response and optimizer configurations. The result retains perceptual and observation parameters, trajectories, likelihood-derived statistics and diagnostics. This organization allows users to examine a trajectory or objective component before accepting a model-selection summary. It also supports existing consumers that rely on MATLAB-style result names.

The accompanying worked-example runner uses the existing fixed 24-trial quickstart inputs and responses, records their hashes, verifies package identity and CPU binary64 execution, and exports

fit fields. Its data are synthetic demonstration vectors, not participant measurements. It is separate from the blocked published-study replications. The clean-install execution and its status are recorded with the runner; an installation failure cannot be replaced by running the current checkout and describing it as the released wheel.

A short usage fragment is:

```
import hgfx
result = hgfx.fit_model(responses, inputs)
trajectory = result.traj.mu
bic = float(result.optim.BIC)
```

The complete runner supplies explicit arrays, environment checks and provenance. It was executed outside the checkout with the released wheel and CPU float64. This synthetic smoke check establishes installation and basic use, not reference parity or recovery. Section 5.4 presents the separate empirical-input example; acquisition and environment details are in Supplement S6.

3.2 Compatibility and execution backends

Compatibility mode targets the frozen reference's statistical and numerical behavior. JAX-backed execution provides another path for supported workloads, but backend availability is not a numerical acceptance test. Validation enables JAX float64 and establishes CPU agreement before accelerator checks. A faster implementation would be useful only if its relevant outputs retained their meaning. This article therefore reports backend evidence as scoped objective agreement, without inferring throughput from the use of JAX or the visibility of a GPU.

The frozen oracle and recorded component fixtures support comparisons after MATLAB is no longer available during routine use. This does not eliminate the need for new reference evidence when a custom response or update equation is introduced. A custom model must specify its inputs, equations, transforms, branches, masks and expected outputs. Passing an unsupported analysis through a similarly named standard function does not establish compatibility.

Figure 1 places the oracle alongside the user workflow. The user fits a perceptual/response pair and inspects results. The oracle supplies evidence about that computation; it is not a required runtime service or an independent source of scientific validity. This distinction avoids conflating reproducible software behavior with an experimentally adequate model.

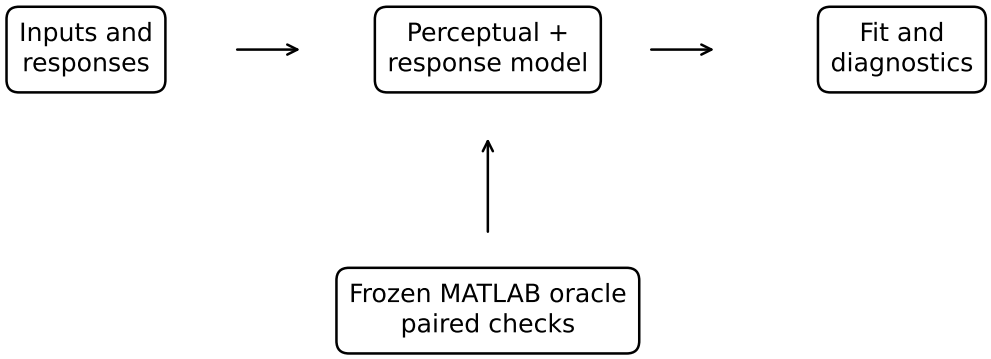


Fig. 1 User workflow and reference checks. Inputs and responses enter a configured perceptual/response pair, followed by fitting and inspection of trajectories, parameters and diagnostics. The separate frozen MATLAB oracle supplies paired evidence. This conceptual schematic contains no measured quantities

4 Evaluation methods

4.1 Frozen scope and evidence interpretation

All reported scientific results concern HGFX 1.0.0 and HGF Toolbox 8.2.0 at the identities in Supplement S1. The revision does not rerun recovery experiments or modify frozen data, seeds, thresholds or failure classifications. Existing machine-readable outputs are the basis of tables and figures. New source inventories and presentation transformations have their own manifests; they do not replace historical artifacts.

The validation hierarchy starts with configuration and transformed/native parameter semantics, then proceeds to numerical primitives, fixed-parameter trajectories, trial likelihoods, objective components, fitting and statistics, and complete workflows. This ordering helps localize disagreement. A trajectory comparison can reveal a first divergent trial, level and state before optimizer differences obscure the cause. A fixed-vector objective check can distinguish a response or prior mismatch from a fitting-path difference. Fitted-parameter checks are meaningful only after these simpler comparisons have a defensible common surface.

Four outcome classes are used. Direct parity means agreement under the frozen criterion for a stated scope. A matched reference limitation means that the oracle exhibits the corresponding instability or limitation; it is not a scientific recovery success. A preserved failure remains a failed criterion in the evidence record. A not-directly-comparable quantity lacks a defensible common semantic or numerical surface. These classifications are not interchangeable labels for successful execution. In particular, product acceptance of a matched limitation does not turn a failed inference experiment into a pass.

Numerical criteria are field-specific. The two official binary demo workflows use 320 trials and compare release-relevant trajectories and inference states at relative tolerance 5×10^{-11} and absolute tolerance 5×10^{-13}. Fitting, curvature and backend criteria have their own scopes; a single tolerance is not applied across all quantities. No criterion in this revision was relaxed to accommodate a

discrepancy. Correlation alone is insufficient because it can remain high despite offsets, scaling changes or a small number of divergent states.

4.2 Paired recovery and model selection

The paired recovery grid uses three binary perceptual models: classic, enhanced and unbounded HGF. It includes truth scales of 0.15 and 0.35 prior standard deviations, trial counts of 128 and 256, and frozen quasi-Newton settings. There are 72 parameter cases and 36 model-recovery datasets, with three candidate fits per model-recovery dataset. Each model contributes 24 parameter cases, pooling two trial counts, two truth scales and six replicates. Fits start once at resolved transformed prior means, with at most 100 quasi-Newton iterations and no random restarts. The three free parameters are omega2, omega3 and log zeta (zero-based combined-vector positions 12, 13 and 14).

Predeclared scientific criteria are convergence rate at least 0.80, median parameter-recovery correlation at least 0.50, median standardized root mean square error (sRMSE) at most 1.00, and model-recovery balanced accuracy at least 0.50. Parameter recovery asks how well fitted parameters recover simulated truths. Model recovery asks whether model selection recovers the generating model. Cross-implementation winner agreement asks a third question: whether MATLAB and HGFX choose the same candidate on the same data. Figure 3 separates the latter two quantities visually.

For each transformed parameter, sRMSE is its root mean squared estimation error divided by the population standard deviation of its simulated truths across the 24 cases. The recorded implementation substitutes a divisor of one only when that standard deviation is at most 10^{-12}. Correlation is Pearson correlation across cases; the reported median includes finite parameter correlations. Nonconverged but returned fits remain in these metrics; an execution failure makes the model summary incomplete. Convergence means termination at the argument or gradient tolerance in HGFX. MATLAB convergence is inferred from the retained iteration trace and absence of its maximum-reset/maximum-iteration warnings, rather than from successful function return. Supplement S2 gives parameter-wise results and the selection confusion matrix. Historical failures remain archived independently.

4.3 Trial-horizon diagnostic extension

A separate prospective experiment considered 128, 256, 512 and 1024 trials with the same three model families and frozen recovery criteria. Its scientific comparison of 256 versus 1024 trials was conditional on a paired-integrity gate. The shorter and intermediate horizons supplied diagnostics. Invalid simulations and failed fits were retained rather than redrawn. This design matters because selectively replacing invalid cases could change the apparent relation between trial length and recovery.

The integrity gate is logically prior to interpretation of horizon trends. A promising recovery metric for one model or horizon does not override disagreement between implementations elsewhere in the required paired scope. The complete aggregate, diagnostic classifications and provenance are provided in Supplement S3. The revision reports the failed gate rather than treating executed shards as proof of a successful scientific experiment.

4.4 pyhgf comparison

The common-scope comparison pins pyhgf 0.3.2 and uses one fully observed, fixed-parameter three-level binary HGF with 128 trials. Model structure, update rules, initialization, parameters, input

coding, masks, precision and numerical guards were mapped before numerical interpretation. The authorized settings include standard volatility updates, mean-field updates, unit coupling, zero drift and native inverse temperature 48. The comparison does not cover all pyhgf networks, all HGFX families, parameter fitting or arbitrary response models.

Mapped trajectory quantities use absolute tolerance 10^{-10} and relative tolerance 10^{-8}. The candidate response total negative log-likelihood (NLL) surface uses absolute tolerance 10^{-7} and relative tolerance 10^{-8}. A response-probability boundary nonfinite was prospectively assigned to the not-directly-comparable class. It was not repaired after inspection by clipping probabilities. The complete mapping and raw outputs make this limitation distinguishable from a perceptual-trajectory mismatch.

4.5 Backend evidence and published-study feasibility

The backend comparison distinguishes compatibility CPU versus JAX CPU from JAX CPU versus physical GPU. The first has a final-objective gap criterion of 0.10; the second uses 10^{-7}. Retained physical-GPU evidence comes from a recorded pre-release runtime with two visible Tesla T4 devices, while the required fitting cells execute on one tested device. Device enumeration and actual array residency are separate checks. These measurements concern applicability and agreement, without a speed or scaling hypothesis.

The empirical example follows the frozen toolbox's continuous-input demonstration: a two-level classic HGF with the bayes_optimal likelihood evaluates the USD/CHF observations directly, with an empty response vector. Default priors, placeholder resolution and optimizer settings are retained. The complete sequence is used in its recorded order, without generated responses or removed rows. It is a descriptive fit to an empirical series, not an out-of-sample forecasting test. Separate source audits of Pain 2025 (Lacombe-Thibault and Coll 2025) and SPIRL (Hess et al. 2025) assess whether their customized participant pipelines can be transferred without changing the models; neither proceeds to fitting.

4.6 Forward validation on published behavioral inputs

To evaluate fixed-parameter agreement on published behavioral inputs, we used two 80-trial blocks or condition halves per participant prefix from two archived behavioral input files (Reed et al. 2020; Suthaharan et al. 2021). The initial 326-block source check stopped before full-cohort model execution because one Reed workspace variable was not a participant identifier. The revised protocol, recorded before the paired full-cohort execution, retained all 260 Suthaharan and 64 Reed blocks in the two pinned source files. We preserved binary outcomes, three-choice coding and the mask for trials with a missing choice or outcome. Both implementations evaluated the same uncoupled three-arm perceptual model at the native 17-parameter vector inherited from reference test M12B. We compared 12 trajectory or inference-state outputs per block in CPU binary64 with relative tolerance 2×10^{-9} and absolute tolerance 2×10^{-11}, including matched nonfinite values. The workflow held all parameters fixed and omitted the observation model. Supplement S7 records the source, protocol and software identities.

5 Results

5.1 Reference workflows and sensitive fits

The official workflow evidence supports scoped reproduction of the binary enhanced-HGF and unbounded-HGF autoregressive workflows. In the documented binary challenge regime, classic HGF

encounters negative posterior precision in both implementations while enhanced HGF completes. Reproducing the expected failure is part of reference behavior. It is not an improvement in the classic model, nor a reason to conceal that case.

Two fitting cases retain endpoint or inference-equivalence failures and are classified as matched reference limitations. The official stress fit is sensitive to optimizer behavior within a numerical basin. The separate fixed-seed holdout fit fails inference equivalence. Their detailed paired records are in Supplement S2. Neither is summarized as direct parameter parity. The core/API, sampling, analysis and Bayesian parameter-averaging evidence concerns the documented release scope; it is not a blanket assertion that all possible scientific workflows are validated.

Table 1. Quantitative evidence and its boundaries. Each criterion applies only to its stated surface. NR means a maximum discrepancy was not retained in the classification artifact; it is not zero

Comparison and scope	Recorded measurement	Frozen criterion / interpretation
Official binary eHGF challenge; 320 trials	No trajectory/state mismatches; maximum NR	atol 5×10^{-13}, rtol 5×10^{-11}; classic HGF fails in both
Official uHGF / AR1 workflow; 320 trials	No trajectory/state mismatches; maximum NR	Same trajectory tolerances; two model paths
Sensitive eHGF fit; shared-state probe	Objective gap 2.84×10^{-14}; state gap zero	Endpoint mismatch preserved; not fit parity
Paired binary model selection	36/36 BIC winners agree	Exact winner identity; recovery accuracy 0.583
Mapped pyhgf perceptual comparison	11 fields; maximum error 1.55×10^{-15}	atol 10^{-10}, rtol 10^{-8}; one 128-trial case
Compatibility / JAX CPU fitting	Maximum objective gap 0.006784	Absolute gap ≤ 0.10; four cells
JAX CPU / T4 fitting	Maximum objective gap 1.42×10^{-14}	Absolute gap $\leq 10^{-7}$; four cells

The trajectory checks, shared-state probe and fitted-objective checks address different quantities. They do not certify arbitrary parameter endpoints or custom study responses. The two sensitive fitting cases remain matched reference limitations; their numerical localization is summarized in Supplement S2.

The software comparison is deliberately asymmetric (Table 2). The MATLAB package is the specification used for this evaluation, HGFX is evaluated as its scoped reimplementation, and pyhgf is compared only after establishing a common surface. Thus a row marked untested describes the limits of this study rather than an absent software capability. Runtime requirements and model semantics should be considered separately: sharing a language or accelerator framework does not make likelihoods or fitted statistics interchangeable. Likewise, retaining MATLAB-style aliases is useful for migration, but interface similarity is weaker evidence than agreement of transformed parameters and objective components. The same rule applies to customized published studies, where familiar perceptual model names can accompany materially different response definitions.

Table 2. Version-bound software comparison. Sources are the frozen oracle, release configuration/API evidence, and the prospective pyhgf semantic mapping. Untested does not mean unsupported

Property	MATLAB HGF 8.2.0	HGFX 1.0.0	pyhgf 0.3.2
Role here	Frozen reference	Reference-compatible Python implementation	Common-scope comparator
User runtime	MATLAB	Python/NumPy/JAX; no MATLAB	Python/JAX
Configuration semantics	Specification	Tested configured surfaces	Mapped in one binary case
Inference	Reference approximate updates	Documented HGF/eHGF/uHGF families	Network-based construction; one mapped cell tested
Fitting comparison here	Paired oracle fits	Direct/limited results retained	Untested
Accelerator evidence here	Untested	Retained T4 objective cells	Untested
Response NLL comparison	Reference in paired HGFX scope	Case-specific	NDC in common-scope cell

5.2 Recovery and implementation agreement

Parameter-recovery criteria are not fully met in the paired grid (Table 3 and Figure 2). Classic and enhanced HGF fail the correlation and sRMSE criteria. Unbounded HGF also fails the convergence criterion. The corresponding MATLAB and HGFX metrics are close, which supports the narrow matched-limitation interpretation; it does not establish that those parameters can be estimated adequately in the tested designs.

Table 3. Paired parameter recovery across both truth scales. Entries are MATLAB/HGFX; displayed values are rounded from full-precision artifacts. Each model contributes 24 parameter cases. Thresholds: convergence >=0.80, median correlation >=0.50, median sRMSE <=1.00

Model	Convergence	Median correlation	Median sRMSE	Failed criteria
Classic HGF	0.833/0.833	0.203/0.203	2.610/2.610	Correlation, sRMSE
Enhanced HGF	0.917/0.917	0.462/0.462	2.359/2.359	Correlation, sRMSE
Unbounded HGF	0.792/0.792	0.381/0.381	2.958/2.958	All three

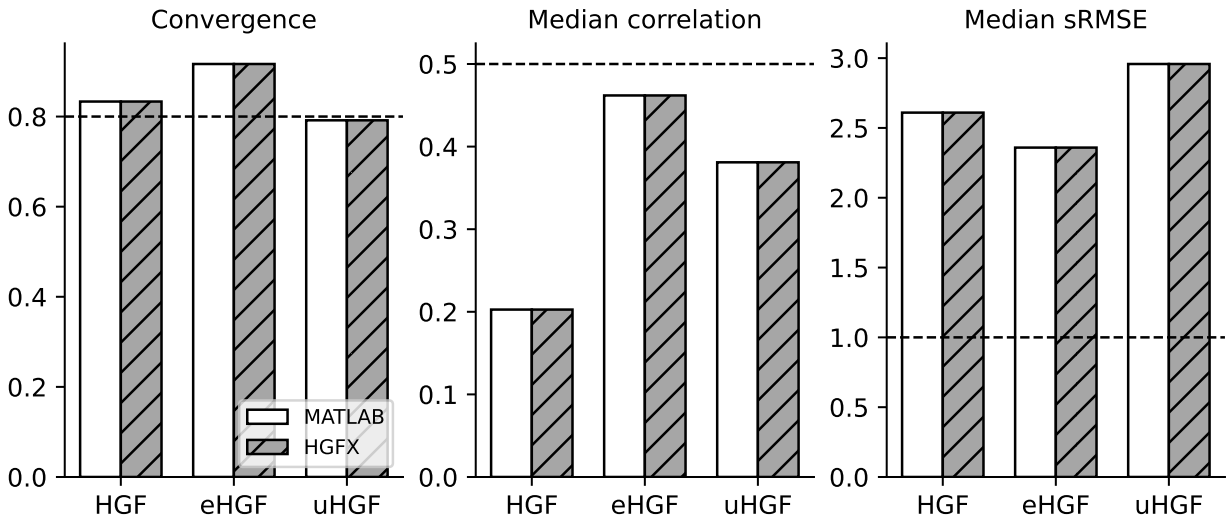


Fig. 2 Paired parameter-recovery summaries for the frozen grid, with the predeclared thresholds shown as dashed lines. These are descriptive aggregate metrics without error bars, not a scientific recovery pass. All three model families retain failed criteria

All 36 model-recovery datasets have the same minimum-BIC winner in MATLAB and HGFX (Figure 3a). Generating-model recovery balanced accuracy is 0.5833333333333334 for both (Figure 3b). The perfect implementation agreement and imperfect generating-model recovery coexist because they measure different properties. The recovery threshold applies to balanced accuracy, not to cross-implementation agreement. This result supports reproducibility of the tested selection behavior, not universal discrimination among HGF families.

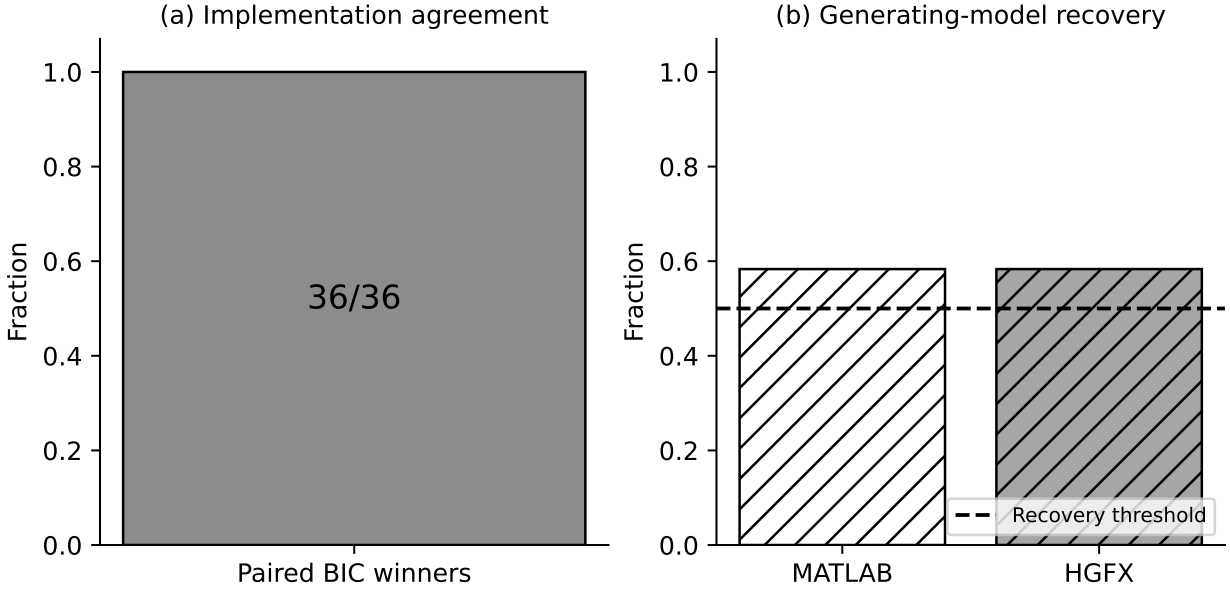


Fig. 3 Implementation agreement and generating-model recovery in the same 36 paired datasets, with three candidate models each. Panel (a) shows 36/36 matching BIC winners. Panel (b) shows balanced accuracy of 0.583 for each implementation; only this panel includes the frozen recovery threshold of 0.50. Bars are deterministic summaries of the frozen grid, without error bars

5.3 Trial horizon, software comparison and backends

The trial-horizon experiment fails its paired-integrity gate. A retrospective audit of the original 24 shard archives clarifies why: all 62 datasets with nonempty selected-model labels agree, while ten datasets have no selected model in either implementation because simulation was invalid. The frozen checker counts these empty pairs as failed winner matches; they are not ten pairs choosing different models. Of 144 parameter cases, 20 share an invalid simulation and four have an HGFX initial-objective rejection with a MATLAB success flag. Two of those MATLAB outputs have nonfinite negative log joint values; the other two return finite values. All 120 successful pairs have the expected free indices. Thus the 24 failed index checks reflect empty failure outputs, not observed parameter-order differences between successful fits. The classifier propagates its global gate failure to all three families, so its labels do not localize a defect to every family. These distinctions correct interpretation without changing the frozen gate, replacing failed data or claiming identifiability. Supplement S3 retains the four asymmetric cases and the limits of their localization.

All 11 mapped perceptual quantities pass the common-scope pyhgf criteria. Maximum absolute differences range from approximately 1.11×10^{-16} to 1.55×10^{-15} in the frozen cell. Participant-response NLL is NDC: the power-ratio transformation on the pyhgf side reaches an exact probability boundary for the recorded inverse temperature, yielding a nonfinite comparison surface. This outcome limits a response comparison while leaving the mapped perceptual agreement intact. Figure 4 shows overlapping trajectories; the field-wise error record in the supplement is necessary to interpret that overlap.

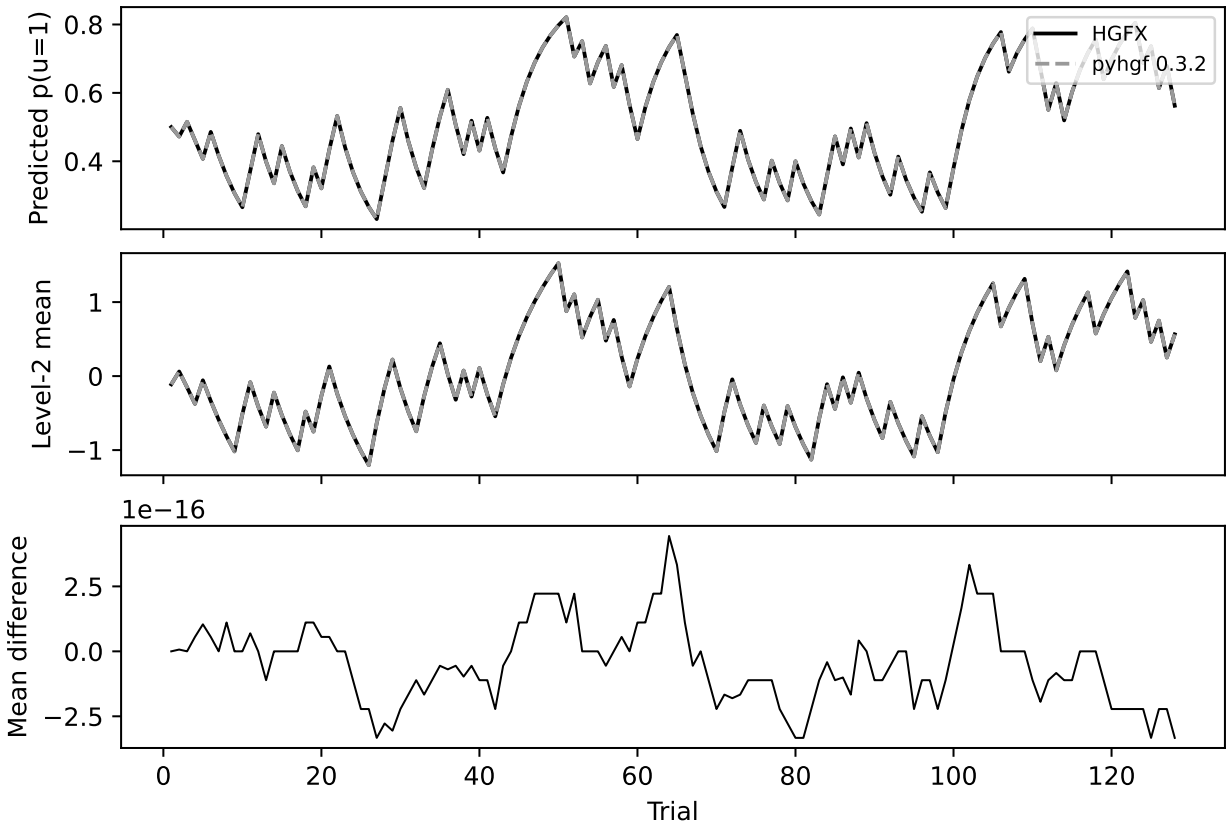


Fig. 4 HGFX and pyhgf 0.3.2 trajectories in the single authorized 128-trial binary-HGF case. Solid and dashed lines distinguish implementations. The first two panels show predicted input probability and level-2 posterior mean; the third shows the signed HGFX-minus-pyhgf level-2 mean difference over all trials. Eleven mapped perceptual quantities pass their criteria. Participant-response NLL remains NDC. No uncertainty bands are shown because these are deterministic fixed-parameter trajectories

The compatibility-versus-JAX CPU maximum final-objective gap is 0.006783711260709424 against its 0.10 criterion. The four retained CPU-versus-T4 fitting cells pass their 10^{-7} criterion, with maximum gap $1.4210854715202004 \times 10^{-14}$. Figure S2 expresses each gap relative to its own threshold. The retained GPU runtime and source identity are provided in Supplement S4. These are different comparisons with different criteria; their gap magnitudes should not be read as a ranking of hardware accuracy.

5.4 Empirical-input workflow and application boundary

The continuous-input example applies the released wheel to the entire USD/CHF series distributed with the frozen toolbox. The upstream demo describes exchange rates during 2010 and 2011; the file supplies values without calendar labels. Consequently, the example uses observation number rather than invented dates and treats the series' upstream description as its provenance. It estimates perceptual parameters from the input likelihood, without participant choices or synthetic response measurements. The run completed in 182 seconds on CPU float64, returning negative log likelihood -2387.622 and BIC -4736.724 for 614 observations. A continuous density can exceed one, so a negative negative-log-likelihood is not itself an error. Figure 5 shows the observations and fitted-filter

predictions. Parameters were estimated from the full series: the one-step predictions are therefore in-sample diagnostics, not held-out forecasts. Numerical-derivative warnings occurred; finite statistics are not treated as a convergence or curvature-validation certificate. Supplement S6 records the execution and remaining diagnostics.

The example shows how an analyst can inspect one-step predictions and changing latent volatility using a standard configuration. It does not establish financial forecasting performance or reproduce a participant study. Pain's custom response permissions and historical environment remain unresolved. SPIRL requires a separately validated custom perceptual model and seven joint choice/response-time models. The source audits identify those boundaries; their file-acquisition details belong to the reproducibility dossier, not to an empirical result.

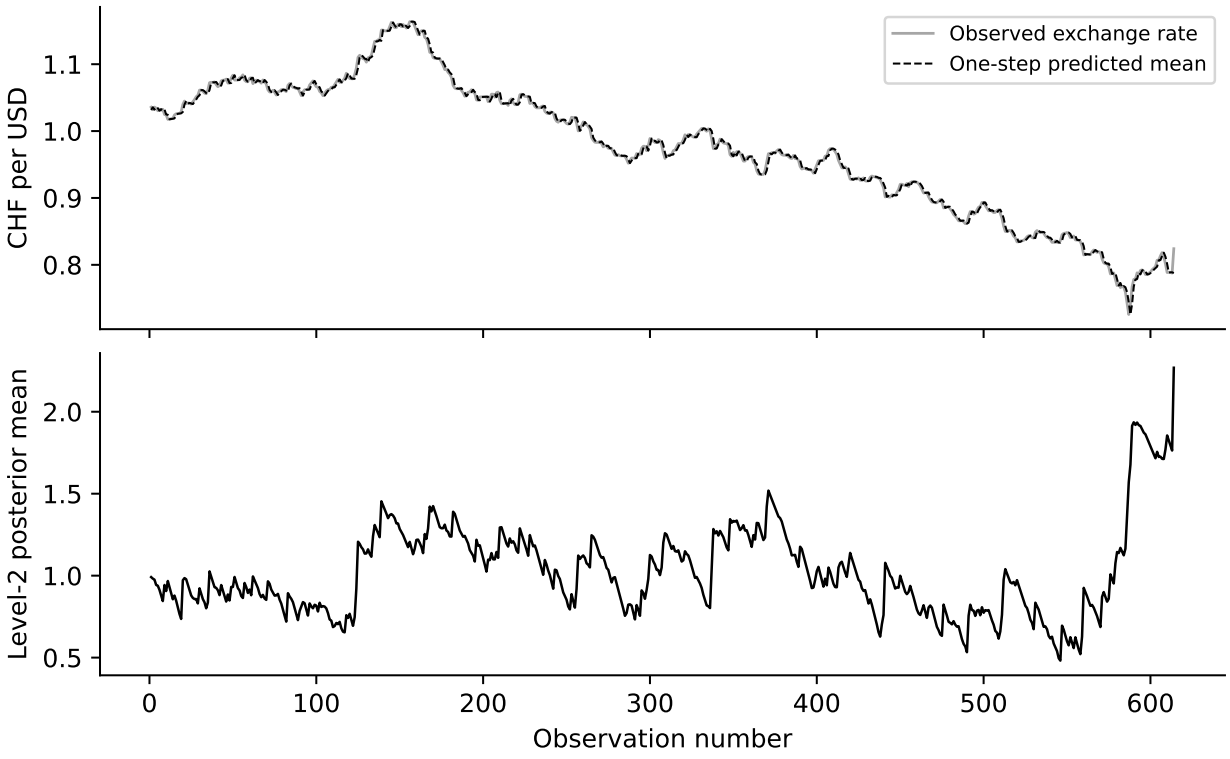


Fig. 5 Empirical-input worked example with all 614 USD/CHF observations from the frozen toolbox demo. The upper panel shows observed rates and one-step predicted means from the continuous two-level HGF; the lower panel shows the fitted level-2 posterior mean, which modulates innovation variance. Parameters are estimated using the full series. No held-out accuracy, parameter uncertainty, new MATLAB agreement or participant-study replication is implied

5.5 Forward agreement on published behavioral inputs

Both implementations completed the fixed-parameter comparison for all 324 blocks (25,920 input trials). Across 3,888 fields and 2,540,160 compared array entries, no entry exceeded the specified criterion. The largest absolute difference among finite pairs was 3.109×10^{-15}. Python emitted warnings in 48 of the 324 blocks. The archived summary retains the warning counts, and the warning mechanisms remain uncharacterized. The result concerns numerical compatibility of the perceptual filter on the two selected input files. Replication of the original fitted parameters, model comparisons and group-level results remains outside this experiment.

6 Discussion

HGFX serves analysts who need the tested conventions of a particular MATLAB HGF release in a Python environment. The contribution is the combination of a usable interface and scoped numerical evidence: parameter spaces, priors, masks, likelihoods and returned statistics are part of the analysis specification. Table 1 separates this evidence into surfaces that a user can evaluate before transporting an existing workflow. A shared model name alone is insufficient.

Implementation agreement and scientific recovery answer different questions. In the paired grid, 36 matching BIC winners coexist with generating-model balanced accuracy of 0.583. Parameter-wise results further show that the median recovery summary conceals substantial heterogeneity: classic HGF recovers omega2 well in this grid, while its omega3 correlation is negative. This does not establish universal failure or success of either parameter. The narrow truth ranges, pooled horizons, observation model and single-start fitting constrain the result. Individual-difference studies need recovery assessment for their own design.

The horizon audit also illustrates why failure labels require inspection of raw outputs. Missing winners after invalid generation are different from competing finite model selections. Empty index arrays after execution failure are different from reordered parameters. Nonetheless, four asymmetric initial-fit outcomes remain, and the historical outputs do not identify their first divergent trial, level and intermediate state. The extended horizon grid therefore remains outside the supported paired-fitting claim. No general structural-identifiability conclusion follows, and the successful subset does not rescue the original gate.

The published behavioral inputs test the same perceptual calculation across observed choice and outcome sequences from two source files. Their numerical agreement extends the synthetic checks to those inputs at one parameter vector. Assessing transportability of the original analyses would require their observation likelihoods, priors, optimization procedures, sample definitions and outcome analyses. The 48 blocks with Python warnings also warrant diagnostic follow-up. The USD/CHF example separately demonstrates fitting for a standard continuous model, while the Pain and SPIRL audits identify open requirements for customized workflows.

A separate untimed diagnostic on 64 synthetic standard-binary HGF datasets further distinguishes forward agreement from complete-fit agreement. Sixty-three fits met the existing full-output criteria. For subject index 7, four level-2 posterior means and four corresponding predicted means exceeded the trajectory criterion, although the fitted parameters and other compared fit quantities met their field-specific criteria. This cohort result limits an end-to-end fitting-equivalence claim. It also motivates comparison of complete fitted trajectories after checking the fixed-parameter calculation. The archived 64-fit record retains the failed values and unchanged thresholds.

HGFX and pyhgf have complementary evaluation scopes here. Eleven mapped perceptual quantities agree at rounding scale in one case, but the response-probability boundary prevents a likelihood-equivalence claim. Likewise, T4 objective agreement does not establish throughput or scaling. Broader response comparisons and performance benchmarks require their own prospective protocols.

Compatibility is tied to the immutable 8.2.0 reference and the tested scopes, not future releases or arbitrary custom models. Two sensitive fitting cases remain exact-scope reference limitations; the horizon failures remain unresolved beyond the recorded diagnostic stages. Within those boundaries, the package makes established HGF workflows accessible without MATLAB at user runtime and gives analysts explicit evidence for deciding what can be transferred and what still requires validation.

7 Data and code availability

HGFX 1.0.0 is available from the versioned release (HGFX developers 2026) under the MIT license. The repository provides the manuscript, evidence manifests and reproducibility instructions. Supplement S7 identifies the original behavioral input files and their source commits, the preparation and comparison code, and the archived numerical summary. Participant inputs and per-trial trajectories are available only through their respective original sources or regeneration under the recorded workflow, as applicable. They are excluded from this package. MATLAB is required to regenerate the reference outputs.

Electronic supplementary material

Online Resource 1 (ESM_1.pdf) contains frozen identities, numerical validation and recovery details, trial-horizon diagnostics, backend and comparator evidence, published-input source provenance, and the bounded warning and feasibility follow-up.

Statements and Declarations

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No specific grant funding was received for this work. Mohammad Ahmadkhanloo is the developer and maintainer of HGFX and declares no other competing interests.

Author contributions and ethics

Mohammad Ahmadkhanloo: conceptualization, software, validation, formal analysis, data curation, visualization, and writing (original draft and review/editing). The software evaluation uses existing publicly archived choice and outcome sequences as numerical inputs and includes an empirical exchange-rate example. It involved no new participant recruitment or data collection. The source studies report their original ethics and consent procedures (Reed et al. 2020; Suthaharan et al. 2021).

Generative AI use and accountability

OpenAI ChatGPT assisted with source inventory, code and test drafting, manuscript restructuring and substantial prose generation, figure preparation, and package checks. This work extends beyond copyediting. Automated source checks and editorial review are recorded as such, not as completed human verification. The author confirmed authorship and declarations and remains responsible for verification and approval of the final scientific content before submission.

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